Competitive Exam Practice 8: Searching and the Elementary Sorts

GATE (Computer Science) Pattern

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October 15, 2026

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. Arrays are 0-indexed unless the question states otherwise.

Q1 *[GATE CSE 1996 Q2.13, verified — GATEOverflow V2 p.95]* **[MCQ, 1 mark]** The average number of key comparisons required for a successful search, for sequential (linear) search on n items, is

(A) $\log n$

(B) $(n-1)/2$

(C) $n/2$

(D) $(n+1)/2$

Q2 *[GATE CSE 2008 Q84, verified — GATEOverflow V2 p.95]* **[MCQ, 2 marks]** Consider the following C program that attempts to locate an element x in an array $Y[\,]$ using binary search. The program is erroneous.

```
f(int Y[10], int x) {
   int i, j, k;
   i = 0; j = 9;
   do {
      k = (i + j) / 2;
      if (Y[k] < x) i = k; else j = k;
   } while ((Y[k] != x) && (i < j));
   if (Y[k] == x) printf(" x is in the array ");
   else printf(" x is not in the array ");
}
```

On which of the following contents of Y and x does the program fail?

(A) Y is [1 2 3 4 5 6 7 8 9 10] and $x < 10$

(B) Y is [1 3 5 7 9 11 13 15 17 19] and $x < 1$

(C) Y is [2 2 2 2 2 2 2 2 2 2] and $x > 2$

(D) Y is [2 4 6 8 10 12 14 16 18 20] and $2 < x < 20$ and x is even

Q3 *[GATE CSE 2006 Q14, verified — GATEOverflow V2 p.101]* **[MCQ, 1 mark]** Which one of the following in-place sorting algorithms needs the minimum number of swaps?

(A) Quick sort

(B) Insertion sort

(C) Selection sort

(D) Heap sort

Q4 *[GATE CSE 2021 Set 1, verified — GATEOverflow V2 p.102]* **[MCQ, 1 mark]** Consider the following array:

$$[23,\ 32,\ 45,\ 69,\ 72,\ 73,\ 89,\ 97].$$

Which algorithm out of the following options uses the least number of comparisons (among the array elements) to sort the above array in ascending order?

(A) Selection sort

(B) Mergesort

(C) Insertion sort

(D) Quicksort using the last element as pivot

Q5 *[GATE CSE 1999 Q1.12, verified — GATEOverflow V2 p.100]* **[MCQ, 1 mark]** A sorting technique is called **stable** if

(A) it takes $O(n \log n)$ time

(B) it maintains the relative order of occurrence of non-distinct elements

(C) it uses the divide and conquer paradigm

(D) it takes $O(n)$ space

Q6 *[Original, GATE-pattern]* **[NAT, 1 mark]** Using the lecture's integer-square-root routine — binary search on the answer space $0..n$ for the largest k with $k^2 \le n$ — the value returned for $n = 90$ is _____. (Answer in integer)

Q7 *[Original, GATE-pattern]* **[MCQ, 2 marks]** The set of strings {`book`, `boot`, `bough`} is kept in sorted order. Using the lecture's fact that the longest common prefix (LCP) of a sorted set equals the common prefix of its first and last strings, what is the LCP of the whole set?

(A) "bo" (length 2)

(B) "boo" (length 3)

(C) "b" (length 1)

(D) the empty string (length 0)

Q8 *[Original, GATE-pattern]* **[MCQ, 2 marks]** The lecture's bubble sort (left to right, with the `swapped` early-exit flag) is run on the array $[4,\ 1,\ 5,\ 2,\ 8]$. After the *first* full pass over the array, what is the array, and how many swaps were performed in that pass?

(A) $[1,\ 4,\ 2,\ 5,\ 8]$ with 2 swaps

(B) $[1,\ 2,\ 4,\ 5,\ 8]$ with 3 swaps

(C) $[1,\ 4,\ 5,\ 2,\ 8]$ with 1 swap

(D) $[1,\ 4,\ 2,\ 5,\ 8]$ with 3 swaps

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Q9 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Consider the lecture's counting sort (tally, prefix sums, then a reverse scatter). Which of the following statements is TRUE?

(A) Counting sort is an in-place comparison sort with $\Theta(n^2)$ worst-case time.

(B) Counting sort runs in $\Theta(n + k)$ time and is stable with the reverse scatter, but it uses $\Theta(n + k)$ auxiliary space and is therefore not in-place.

(C) Counting sort is in-place and unstable.

(D) Counting sort sorts keys of any real-valued range in $\Theta(n)$ time.

Q10 *[Original, GATE-pattern]* **[NAT, 1 mark]** The lecture's insertion sort is run on the array that is already in ascending order: [10, 20, 30, 40, 50, 60]. Counting every evaluation of the condition `A[j] > key`, how many comparisons are performed in total? _____. (Answer in integer)

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